BSc Degree in Applied Statistics

Statistics in Health Sciences

2. PubMed: Searching scientific papers. Structure of a paper

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Statistics in Health Sciences

1. PubMed: Searching scientific papers

2. The scientific paper

Warning



During class sessions, **it is not allowed to:**

- Take photos,
- Record videos,
- Record audio.

PubMed: Searching scientific papers

What is PubMed?

- NLM has been indexing the biomedical literature since 1879, in order to help health professionals in accessing information needed for research, health attendance and education.
- Later, it becomes a huge data base, which nowadays is known as MEDLINE and contains citations and abstracts of biomedical journals worldwide.
- Since 1996, free online access to MEDLINE is available via PubMed.

PubMed: Searching scientific papers

How does PubMed work?

- PubMed is a tool for search of scientific papers. Such search, can be done filtering according to one or several characteristics of the paper such as: author, title, journal, word in the text, etc.
- Tutorials about PubMed are available at:
  http://www.nlm.nih.gov/bsd/disted/pubmedtutorial/
- In the previous tutorial several illustrative videos can also be found, such as:
  http://www.nlm.nih.gov/bsd/disted/pubmed.html#qt

The scientific paper

Characteristics

- The main tool for the publication and divulgation of a biomedical study (and, in general, any scientific study) is the publication of a scientific report in a specialised journal. Such publications are known as (scientific) papers.

- Dispite it can be slight differences according to each journal's publication rules, all papers usually have a similar format.
  - For instance, a typical format include: black and white print, one of two columns, and an extension around 10 pages (although there are no extension restrictions in some journals).
  - The paper usually include some tables or figures, which must include an autoexplicative caption.

The scientific paper

Structure

The paper is usually (but not always) structured as follows:

- Title, authorship and affiliation
- Abstract
- Introduction
- Methods
- Results
- Discussion/Conclusions
- References
- Supplemental material

The scientific paper: Structure

Title, authorship and affiliation

- The title should be short but explicative enough in order to give an idea of the contents.
- The criterium for the order of the authors can vary. Often, the first author is who has written the paper, followed by the remaining coauthors. In the case of studies developed by a working group that has a leader, the leader usually appears in the last position.
- All authors must shown their affiliation, which is the research centre or university they work for.

The scientific paper: Structure

Abstract

- It is a brief summary of the paper.
- It usually takes few lines.
- In some journals, it must label the sections: background, methods, results, discussion/conclusions.
- It usually shows some keywords related to the paper's contents.

The scientific paper: Structure

Introduction

- In this section, the study is introduced.
- Aims: what is the scientific question the researches are interested in?
- Motivation: why the study is interesting? What is the authors' contribution?
- Background: what has previously been done related to the topic?

The scientific paper: Structure

Methods

- It is the most important sections from a statistical point of view.
- In this section, the methods related to data gathering and modelling are detailed:
  - Population: sampling, inclusion/exclusion criteria
  - Study design: randomised clinical trial, case-control, cohort, cross-sectional, ...
  - Information gathering protocol
  - Statistical analysis (methods, software, ...)

The scientific paper: Structure

Results

- In this section, the main results of the statistical analysis (in the Methods section) are shown.
- It should be clear and concise.
- Statistical summary indicators related with the aims of the study are given.
- Such indicators usually include a significance measure, mainly confidence intervals and p-values.
- Results usually are illustrated with (a limited number of) tables and figures.

The scientific paper: Structure

Discussion

- This section can be seen as a "translation" of the results from the "statistical language" to the "epidemiological language", according to the context of the study.
- Results are usually linked to the previous findings that have been shown in the background section, discussing potential incoherences.
- Limitations of the study should be stated as, for instance, issues during the information gathering, missing data, lost of follow-up, low power, etc.
- Finally, ideas on how to improve or continue the investigation are often provided.

The scientific paper: Structure

Supplemental material

- This section include materials and results that are not of principal interest but that illustrate or complement the main results. For instance:
  - Further tables of figures
  - Data
  - Secondary analyses (e.g., sensitivity analyses)

Exercise

Exercise

✎ Use the advanced search of PubMed to find a paper, published not before 2015, that assessed, in some Spanish city, the association between ambient air polution and overweight in school-aged children.

1. Write the syntax you have used to find the paper.

2. Write the following information about the paper:

   - Title
   - Names of the authors and their affiliations
   - Date of publication
   - Name of the journal in which the paper was published

3. Summarize aim, methods and results. Compare your results with those generated by an AI.